A PROJECT REPORT ON
"OPTIMIZING AGRICULTURE USING MACHINE LEARNING AND DEEP LEARNING"

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF SCIENCE (MSDs)

Submitted By

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CERTIFICATE

This is to certify that the project work titled "**OPTIMIZING AGRICULTURE USING MACHINE LEARING AND DEEP LEARNING**" submitted to the **VEDA DEGREE COLLEGE**, affiliated to **OSMANIA UNIVERSITY**, by **MOHAMMED ABDUL JUNAID** (2803-23-539-027), **MOHD ATHEEQ KHAN** (2803-23-539-033), **MOHD YOUSUF** (2803-23-539-034), **SHAIK JUNAID** (2803-23-539-040), **YOUSUF BIN AMER NAHDI** (2803-23-539-049), is a bonafide record of the work done by the students towards partial fulfillment of requirements for the awards of the degree of **Bachelor of Science (MSDs)** during the academic year 2025-2026.

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I hereby declare that this project report on **"IMPROVING AGRICULTURE USING MACHINE LEARNING AND DEEP LEARNING"** has been prepared by me during the month of JAN-MAR, 2026 under the guidance of **Mrs.D.SUCHARITA**, Lecturer, Department of **DATA SCIENCE,** Veda Degree College, Champapet.

I also hereby declare that this project report is an original work and has not been submitted at any time to any other university or institute for the award of any degree or diploma.

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IMPROVING AGRICULTURE USING MACHINE LEARNING AND DEEP LEARNING

ABSTRACT

The proposed system titled **"Optimizing Agriculture using Machine Learning"** is an intelligent agriculture assistance system developed using Python, Machine Learning, Deep Learning, and Image Processing techniques. The system is designed to help farmers and agricultural researchers make accurate decisions regarding soil classification, leaf disease detection, fertilizer recommendation, and crop prediction.

The system integrates Convolutional Neural Networks (CNN), Decision Tree Regression, Principal Component Analysis (PCA), Gray Level Co-occurrence Matrix (GLCM), histogram-based feature extraction, and K-means clustering. Soil images are analyzed to determine soil type and recommend suitable crops. Leaf images are processed and classified to detect plant diseases and suggest appropriate fertilizers. Additionally, environmental parameters such as temperature, humidity, rainfall, pH, and nutrient values are used to predict suitable crops.

The system is implemented using a Tkinter-based Graphical User Interface (GUI), making it interactive and user-friendly. Performance evaluation is carried out using metrics such as Accuracy, Precision, Recall, and F1-Score along with confusion matrix visualization. The proposed system provides an automated, scalable, and efficient solution for modern agricultural decision-making.

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CHAPTER – 1

1. INTRODUCTION

INTRODUCTION

Agriculture is the backbone of many economies and plays a vital role in ensuring food security and sustainable development. The productivity and quality of crops depend on several factors such as soil type, climatic conditions, nutrient content, and plant health. In traditional farming practices, decisions related to soil classification, crop selection, and disease identification are primarily based on manual inspection and farmer experience. While these methods have been followed for decades, they often lack precision, scalability, and efficiency.

With the rapid growth of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), modern agriculture is gradually transforming into smart agriculture. Machine learning models can analyze large volumes of agricultural data and provide accurate predictions. Image processing techniques can detect diseases in plant leaves at early stages. Predictive models can recommend suitable crops based on environmental conditions. These technologies help reduce human effort, minimize financial losses, and improve overall crop yield.

The proposed system, "Optimizing Agriculture using Machine Learning," integrates multiple machine learning algorithms and image processing techniques to automate soil classification, leaf disease detection, fertilizer recommendation, and crop prediction. The system is developed using Python and implemented with a user-friendly Graphical User Interface (GUI) using Tkinter. It provides farmers with an intelligent decision-support tool that enhances productivity and sustainability.

1.1 Background

Agricultural productivity is directly influenced by soil characteristics such as texture, structure, moisture content, and nutrient availability. Different crops grow best in specific soil types. For example, alluvial soil supports crops like rice and wheat, while black soil is suitable for cotton cultivation. Traditionally, soil testing is conducted in laboratories to determine soil type and fertility. This process is time-consuming, expensive, and not easily accessible in rural areas.

Similarly, plant diseases significantly affect crop yield and quality. Leaf diseases such as blight, rust, and fungal infections cause major agricultural losses worldwide. Manual disease detection through visual inspection may lead to misclassification because symptoms of different diseases often appear similar. Early detection is crucial to prevent disease spread and reduce economic damage.

Crop prediction is another critical aspect of agriculture. Farmers need to decide which crop to grow based on environmental parameters such as temperature, humidity, rainfall, and soil nutrients (Nitrogen, Phosphorus, Potassium). Making incorrect decisions may result in low yield and financial loss.

With advancements in Machine Learning and Deep Learning, automated systems can now analyze images and environmental data efficiently. Convolutional Neural Networks (CNN) are widely used for image classification tasks such as soil and leaf analysis. Decision Tree algorithms are effective for structured data prediction such as crop recommendation. These technologies provide faster and more accurate agricultural solutions.

1.2 Motivation

Farmers often face challenges such as unpredictable weather conditions, improper fertilizer usage, soil misclassification, and delayed disease detection. Many rural farmers do not have access to advanced laboratory testing facilities or agricultural experts. As a result, they rely on traditional knowledge and assumptions, which may not always be accurate.

The motivation behind this project is to develop a smart agricultural system that uses Machine Learning and Image Processing to assist farmers in making data-driven decisions. By automating soil classification and disease detection, the system reduces manual effort and improves accuracy. The integration of crop prediction and fertilizer recommendation further enhances agricultural productivity.

The goal is to create an intelligent, cost-effective, and easy-to-use system that supports farmers in selecting the right crop, identifying diseases early, and applying suitable fertilizers at the right time.

1.3 Problem Statement

Traditional agricultural practices rely heavily on manual inspection and experience-based decision-making. Soil classification requires laboratory testing, which is time-consuming and costly. Leaf disease detection depends on expert observation, which may lead to errors. Crop selection without proper analysis of environmental parameters can reduce productivity.

There is a need for an automated agricultural assistance system that can:

• Classify soil types using image processing techniques.

• Detect and classify plant leaf diseases accurately.

• Recommend suitable fertilizers based on detected diseases.

• Predict appropriate crops based on environmental and nutrient parameters.

The system should provide accurate predictions, reduce human intervention, and improve agricultural efficiency.

1.4 Aim

The main aim of this project is to design and develop an intelligent agricultural optimization system using Machine Learning, Deep Learning, and Image Processing techniques to automate soil classification, disease detection, fertilizer recommendation, and crop prediction.

:

1.5 Objectives

1. To extract texture and statistical features from soil images using GLCM and histogram techniques.
2. To classify soil types using Convolutional Neural Networks.
3. To segment infected regions in leaf images using K-means clustering and HSV masking.
4. To detect and classify plant diseases using CNN models.
5. To recommend appropriate fertilizers based on detected disease.
6. To predict suitable crops using Decision Tree Regression based on environmental parameters.
7. To evaluate model performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

1.6 Scope of the Project

The scope of this project includes agricultural image analysis and predictive modeling. The system can be used as a decision-support tool for farmers, agricultural researchers, and agricultural institutions. It can be further extended to include mobile application deployment, cloud integration, and IoT-based real-time environmental data monitoring.

The project focuses on selected soil types, leaf diseases, and crop categories. Future enhancements can expand the dataset and improve model performance by incorporating more advanced deep learning architectures.

CHAPTER - 2

2. LITERATURE SURVEY

2.1 LITERATURE SURVEY

[1] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep learning in agriculture: A survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90.

This survey paper provides an extensive overview of deep learning applications in modern agriculture. The authors systematically analyze how advanced neural network architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Deep Belief Networks are applied to agricultural tasks including plant disease detection, soil classification, crop type recognition, weed detection, and yield prediction. The study emphasizes that deep learning models significantly outperform traditional machine learning algorithms because they automatically extract hierarchical features from raw data without requiring manual feature engineering. The paper also highlights the growing importance of image-based agricultural diagnostics enabled by affordable cameras and smartphones.

Furthermore, the authors discuss practical challenges in implementing deep learning systems in agricultural environments. These include limited annotated datasets, computational complexity, overfitting issues, and environmental variations such as lighting and background noise. The paper suggests solutions such as data augmentation, transfer learning, and cloud-based deployment. This survey strongly supports the adoption of CNN-based classification systems for agricultural image analysis, forming a theoretical foundation for intelligent crop and soil prediction systems.

[2] S. Sladojevic et al., "Deep neural networks based recognition of plant diseases by leaf image classification," *Computational Intelligence and Neuroscience*, vol. 2016.

This research presents a deep neural network framework for automatic recognition of plant diseases using leaf images. The authors collected a dataset containing healthy and infected leaves and trained a CNN model capable of classifying multiple disease categories. The model architecture includes convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification. The results demonstrate significantly higher accuracy compared to traditional approaches such as Support Vector Machines and K-Nearest Neighbors.

In addition to classification performance, the paper discusses preprocessing strategies such as image resizing, background removal, and normalization to enhance model robustness. The authors highlight that deep learning eliminates the need for handcrafted features, making the system scalable and adaptable to different crops. The study proves that automated disease detection systems can support farmers by providing early and accurate diagnosis, reducing dependency on agricultural experts.

[3] M. Mohanty, D. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7.

This study explores the effectiveness of deep convolutional neural networks in classifying plant diseases using a publicly available dataset. The authors utilize transfer learning techniques with pre-trained architectures such as AlexNet and GoogleNet to improve classification accuracy. The results show that transfer learning significantly enhances performance, especially when the dataset size is limited. The research demonstrates that deep networks can achieve high accuracy even under complex visual conditions.

Moreover, the authors evaluate model performance under varying lighting and environmental backgrounds to simulate real-world agricultural conditions. The findings indicate that deep learning models maintain robustness despite variations in image quality. The study emphasizes the importance of AI-driven disease detection systems for sustainable agriculture and increased crop productivity.

[4] R. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Computers and Electronics in Agriculture*, vol. 145.

This paper evaluates several deep learning architectures for plant disease detection and compares their classification performance. The author tests multiple CNN models and analyzes their strengths and limitations. The study demonstrates that deeper architectures achieve better feature representation and higher accuracy but require careful tuning to prevent overfitting.

Additionally, the research discusses the use of dropout layers, regularization techniques, and optimization algorithms to enhance model generalization. The findings confirm that deep learning-based systems can reach near-human-level accuracy in disease diagnosis. This research validates the practicality of deploying CNN-based classification models in agricultural environments.

[5] J. G. A. Barbedo, "Factors influencing the use of deep learning for plant disease recognition," *Biosystems Engineering*, vol. 172.

This paper investigates various factors that influence the performance of deep learning models in plant disease recognition. The author analyzes dataset size, image resolution, class imbalance, and environmental variability. The study concludes that dataset diversity plays a critical role in improving generalization capability.

The research also discusses preprocessing techniques such as segmentation and color space transformation to isolate infected regions. The findings emphasize that proper dataset preparation significantly enhances classification accuracy. These insights are directly relevant to systems that combine preprocessing and CNN-based classification for agricultural applications.

[6] H. Alsharif et al., "Soil classification using convolutional neural networks," *IEEE Access*, vol. 8.

This research proposes a convolutional neural network model for classifying soil types using image data. The authors collect soil samples and train a CNN architecture to identify different soil categories based on texture and color features. The results demonstrate improved performance compared to classical machine learning algorithms.

The study highlights that deep learning models effectively capture complex spatial patterns in soil images. It also discusses the importance of normalization and augmentation to improve generalization. The research supports the use of image-based soil classification systems for agricultural decision-making.

[7] S. Ramesh and D. Vydeki, "Recognition and classification of paddy leaf diseases using deep neural networks," *Computers and Electronics in Agriculture*, vol. 165.

This paper presents a deep neural network framework for paddy leaf disease detection. The authors implement convolutional layers for hierarchical feature extraction and achieve high classification accuracy. The system effectively differentiates between healthy and diseased leaves.

The study emphasizes the role of automated disease detection in reducing economic losses in agriculture. It demonstrates that deep learning systems can assist farmers in identifying infections at early stages, promoting timely treatment and increased yield.

[8] T. Rahman et al., "Transfer learning with deep convolutional neural networks for plant disease detection," *IEEE Access*, vol. 8.

This study applies transfer learning techniques to enhance plant disease detection accuracy. By fine-tuning pre-trained CNN models, the authors achieve faster convergence and improved performance.
The research highlights reduced computational cost and improved generalization, making transfer learning suitable for real-time agricultural systems. It supports the integration of optimized deep learning models in smart farming applications.

[9] L. Too et al., "A comparative study of fine-tuning deep learning models for plant disease identification," *Computers and Electronics in Agriculture*, vol. 161.

This paper compares multiple deep learning architectures for plant disease identification. The authors analyze performance metrics such as accuracy, precision, recall, and F1-score.
The findings reveal that fine-tuned models significantly outperform baseline networks. The study emphasizes model optimization as a key factor in achieving reliable agricultural classification systems.
[10] M. Brahimi et al., "Deep learning for tomato disease detection," *Applied Artificial Intelligence*, vol. 32.

This research develops a CNN-based model specifically for tomato leaf disease detection. The authors incorporate segmentation and classification steps to enhance performance.
The study confirms that automated disease detection improves productivity and reduces manual labor. It highlights the importance of integrating AI technologies into agricultural practices.

2.2 EXISTING METHODS

Before the implementation of deep learning-based intelligent agricultural systems, several traditional and conventional methods were widely used for plant disease detection, soil classification, and crop

recommendation. These methods mainly relied on manual inspection, basic image processing techniques, and classical machine learning algorithms. However, these approaches were not used in the current project because they have several limitations in terms of accuracy, scalability, and automation.

Traditional Manual Inspection

In conventional agricultural practices, farmers and agricultural experts visually inspect plant leaves and soil samples to identify diseases or determine soil type. Diagnosis is performed based on visible symptoms such as color change, spots, texture variation, or dryness. Soil analysis is often conducted through laboratory testing or physical observation of texture and color.

Image Processing with Handcrafted Feature Extraction

Earlier systems used traditional image processing techniques such as color histogram analysis, texture analysis using Gray Level Co-occurrence Matrix (GLCM), edge detection, and threshold-based segmentation. After extracting handcrafted features, classifiers like K-Nearest Neighbor (KNN) or Naïve Bayes were applied for classification.

Support Vector Machine (SVM) Based Classification

Support Vector Machine was widely used for plant disease classification and soil prediction. In this approach, features were manually extracted from images and then fed into the SVM classifier to separate classes using hyperplanes.

K-Nearest Neighbor (KNN) Algorithm

KNN is a distance-based classification algorithm used in agricultural prediction systems. It classifies input data based on similarity with neighboring training samples using distance metrics like Euclidean distance.

Decision Tree and Random Forest Methods

Decision Tree and Random Forest algorithms were used for crop recommendation and soil classification using structured data such as pH value, moisture, temperature, and nutrient levels. These algorithms work by splitting data into branches based on feature importance.

Artificial Neural Networks (ANN) without Deep Architecture

Earlier systems used shallow Artificial Neural Networks with one or two hidden layers. These networks required manual feature extraction before classification and were limited in handling complex image data.

Disadvantages of Existing Methods

The traditional and classical methods discussed above have several limitations which reduce their effectiveness in modern intelligent agricultural systems.

Manual inspection is time-consuming and depends heavily on expert knowledge. It is prone to human error and inconsistency, especially when dealing with large-scale farms. It also lacks automation and cannot provide real-time analysis.

Image processing with handcrafted feature extraction requires domain expertise to design suitable features. These features may not generalize well under different lighting, background, and environmental conditions. The performance of such systems significantly decreases when images vary in quality.

Support Vector Machine and KNN classifiers perform well only when features are properly engineered. They struggle with large datasets and high-dimensional image data. Additionally, they cannot automatically learn hierarchical features like deep learning models.

Decision Tree and Random Forest methods are more suitable for structured numerical data rather than complex image datasets. They may suffer from overfitting and may not capture deep spatial patterns present in images.

Shallow Artificial Neural Networks have limited learning capacity compared to deep convolutional networks. They are unable to effectively extract spatial features from images, leading to lower accuracy in image-based classification tasks.

Overall, the major disadvantages of existing methods include lower accuracy, dependency on manual feature extraction, lack of scalability, sensitivity to environmental variations, and limited automation capability. These limitations motivated the adoption of a deep learning-based proposed method in the current project.

2.3 PROPOSED METHOD

The proposed system is a Machine Learning–based agricultural optimization system developed using Python with a Tkinter graphical interface. The system integrates image processing, feature extraction techniques, dimensionality reduction, deep learning models, and classical machine learning algorithms to perform soil classification, plant disease detection, and crop recommendation. Unlike traditional systems that rely only on raw images or manual observation, the proposed method combines texture analysis, segmentation, PCA, CNN, and Decision Tree algorithms to improve prediction accuracy and automation. The proposed method consists of three major modules:

Soil Classification Using Texture Features and CNN

In this module, soil images are used to classify soil types such as Alluvial soil, Black soil, Clay soil, and Red soil. The dataset is organized class-wise and images are resized to 64×64 pixels for uniform processing.

After resizing, advanced feature extraction is performed. Texture features are extracted using Gray Level Co-occurrence Matrix (GLCM), which captures spatial relationships between pixel intensities. From the GLCM matrix, properties such as contrast, correlation, energy, homogeneity, and dissimilarity are computed. In addition to texture features, histogram-based color features and statistical measures such as mean, standard deviation, entropy, kurtosis, and skewness are extracted. These combined features effectively represent soil texture and color characteristics.

The extracted feature vectors are normalized and reshaped before being passed into a Convolutional Neural Network. The CNN model consists of convolution layers, max pooling layers, flatten layer, and dense layers with softmax activation for multi-class classification. The model is trained using categorical cross-entropy loss and Adam optimizer. After training, the model predicts the soil type and recommends suitable crops based on the classified soil.

Leaf Disease Detection Using Segmentation, PCA and CNN

This module detects plant leaf diseases from input images. Initially, the leaf image is resized and converted into HSV color space. Green masking is applied to remove healthy leaf portions so that infected regions can be isolated. K-means clustering is also applied for segmentation to separate the diseased area from the background.

After segmentation, the image is flattened into feature vectors. Since raw image features are high-dimensional, Principal Component Analysis (PCA) is applied to reduce dimensionality while retaining important information. PCA helps in reducing computational complexity and improving model performance.

The reduced feature set is then given to a CNN model. The CNN contains dense layers with ReLU activation and dropout layers to prevent overfitting. The final softmax layer classifies the disease into predefined categories. Based on the predicted disease, the system also recommends an appropriate fertilizer to assist farmers in treatment decisions.

Crop Recommendation Using Decision Tree Algorithm

The crop recommendation module works on structured environmental and soil parameter data. The features used include Temperature, Humidity, pH value, Rainfall, Nitrogen, Phosphorus, and Potassium levels.

The dataset is split into training and testing sets. A Decision Tree Regressor is trained using these input features to learn relationships between environmental conditions and crop types. Decision Tree is chosen because it can handle nonlinear relationships and provides clear decision boundaries.

When a user enters environmental values through the graphical interface, the trained Decision Tree model predicts the most suitable crop. This helps in selecting crops scientifically based on soil nutrients and climatic conditions.

Overall, the proposed method integrates texture analysis, image segmentation, dimensionality reduction, deep learning, and machine learning into a unified intelligent agricultural support system. This hybrid architecture improves prediction accuracy, reduces manual effort, and enhances decision-making in agriculture.

Advantages

The proposed agricultural optimization system offers several significant advantages over traditional and single-algorithm approaches. By integrating image processing, texture feature extraction, dimensionality reduction, deep learning, and machine learning models into one unified framework, the system provides accurate, automated, and reliable agricultural decision support.

One of the major advantages is improved classification accuracy. The combination of GLCM texture features, histogram-based statistical features, and CNN enables better representation of soil patterns and disease characteristics. Instead of relying on raw pixel values alone, the system extracts meaningful texture and statistical properties before classification. This improves the discriminative capability of the model, especially when soil types or diseases appear visually similar.

Another important advantage is automated disease detection. The system uses HSV-based green masking and K-means clustering to isolate infected regions from healthy parts of the leaf. This targeted segmentation ensures that the classifier focuses only on relevant disease-affected areas. As a result, the prediction becomes more reliable and less sensitive to background noise.

Dimensionality reduction using Principal Component Analysis significantly reduces computational complexity. High-dimensional image data is compressed into a smaller feature space without losing

essential information. This reduces training time, improves processing speed, and enhances model efficiency while maintaining performance.

The use of Convolutional Neural Networks provides strong feature learning capability. CNN automatically learns complex patterns, edges, and textures that may not be captured by traditional algorithms. Dropout layers help reduce overfitting, improving generalization on unseen data. The model is therefore more robust to variations in lighting, orientation, and image quality.

The crop recommendation module adds practical agricultural value. By using environmental parameters such as temperature, humidity, pH, rainfall, and nutrient levels (N, P, K), the Decision Tree model predicts suitable crops scientifically. This supports farmers in making data-driven crop selection decisions rather than relying solely on traditional practices.

The system provides an integrated multi-module solution. Soil classification, disease detection, fertilizer recommendation, and crop prediction are all combined within a single platform. This reduces the need for multiple independent systems and makes agricultural analysis more convenient and efficient.

The graphical user interface developed using Tkinter makes the system user-friendly. Users can upload images or enter environmental parameters without technical knowledge. This improves accessibility for farmers and agricultural practitioners.

The system is scalable and extensible. New soil types, disease categories, or crops can be added by retraining the model with additional data. The architecture allows future improvements such as adding more advanced deep learning models or integrating cloud deployment.

Overall, the proposed system enhances accuracy, automation, computational efficiency, scalability, and usability. It provides a comprehensive and intelligent agricultural decision support framework that improves productivity and supports sustainable farming practices.

CHAPTER – 3

3. SYSTEM REQUIREMENTS

3.1 SOFTWARE REQUIREMENTS

• Operating System: Windows 10 or above / Linux / macOS
• Programming Language: Python 3.7 or above
• GUI Framework: Tkinter
• Image Processing Library: OpenCV (cv2)
• Numerical Computation: NumPy
• Data Handling: Pandas
• Machine Learning Library: Scikit-learn
• Deep Learning Framework: TensorFlow and Keras
• Feature Extraction: Scikit-image (GLCM features)
• Data Visualization: Matplotlib and Seaborn
• Image Handling: Pillow (PIL)
• Development Tools: Anaconda / PyCharm / VS Code
• Package Manager: pip or Anaconda

3.2 HARDWARE REQUIREMENTS

• Processor: Intel Core i3 or higher (Recommended: Intel Core i5 or above)
• RAM: Minimum 4 GB (Recommended: 8 GB or higher)
• Hard Disk: Minimum 10 GB free storage
• Graphics: Standard integrated graphics (Optional: NVIDIA GPU for faster training)
• Display: Minimum 1366 × 768 resolution
• Input Devices: Keyboard and Mouse

CHAPTER – 4

4. METHODOLOGY

4.1 Methodology: System Architecture

The proposed system integrates Convolutional Neural Networks (CNN), Decision Tree Regression, Principal Component Analysis (PCA), Gray Level Co-occurrence Matrix (GLCM), histogram-based feature extraction, and K-means clustering. Soil images are analyzed to determine soil type and recommend suitable crops. Leaf images are processed and classified to detect plant diseases and suggest appropriate fertilizers. Additionally, environmental parameters such as temperature, humidity, rainfall, pH, and nutrient values are used to predict suitable crops.

The system is implemented using a Tkinter-based Graphical User Interface (GUI), making it interactive and user-friendly. Performance evaluation is carried out using metrics such as Accuracy, Precision, Recall, and F1-Score along with confusion matrix visualization. The proposed system provides an automated, scalable, and efficient solution for modern agricultural decision-making.

4.2 Methodology: Soil Classification Using Texture Features and CNN

Feature Extraction Process

In this module, soil images are used to classify soil types such as Alluvial soil, Black soil, Clay soil, and Red soil. The dataset is organized class-wise and images are resized to 64×64 pixels for uniform processing.

After resizing, advanced feature extraction is performed. Texture features are extracted using Gray Level Co-occurrence Matrix (GLCM), which captures spatial relationships between pixel intensities. From the GLCM matrix, properties such as contrast, correlation, energy, homogeneity, and dissimilarity are computed.

CNN Classification

In addition to texture features, histogram-based color features and statistical measures such as mean, standard deviation, entropy, kurtosis, and skewness are extracted. These combined features effectively represent soil texture and color characteristics.

The extracted feature vectors are normalized and reshaped before being passed into a Convolutional Neural Network. The CNN model consists of convolution layers, max pooling layers, flatten layer, and dense layers with softmax activation for multi-class classification.

The model is trained using categorical cross-entropy loss and Adam optimizer. After training, the model predicts the soil type and recommends suitable crops based on the classified soil.

4.3 Methodology: Leaf Disease Detection Using Segmentation, PCA and CNN

Image Preprocessing

The leaf image is resized and converted into HSV color space. Green masking is applied to remove healthy leaf portions so that infected regions can be isolated.

Image Segmentation

K-means clustering is applied for segmentation to separate the diseased area from the background. After segmentation, the image is flattened into feature vectors.

Dimensionality Reduction

Since raw image features are high-dimensional, Principal Component Analysis (PCA) is applied to reduce dimensionality while retaining important information.

CNN Classification

The reduced feature set is then given to a CNN model. The CNN contains dense layers with ReLU activation and dropout layers to prevent overfitting. The final softmax layer classifies the disease into predefined categories.

Based on the predicted disease, the system also recommends an appropriate fertilizer to assist farmers in treatment decisions.

4.4 Methodology: Crop Recommendation Using Decision Tree Algorithm

The crop recommendation module works on structured environmental and soil parameter data. The features used include Temperature, Humidity, pH value, Rainfall, Nitrogen, Phosphorus, and Potassium levels.

Dataset Preparation

The dataset is split into training and testing sets. A Decision Tree Regressor is trained using these input features to learn relationships between environmental conditions and crop types.

Model Training

Decision Tree is chosen because it can handle nonlinear relationships and provides clear decision boundaries. The model learns patterns from historical agricultural data.

Prediction

When a user enters environmental values through the graphical interface, the trained Decision Tree model predicts the most suitable crop.

This helps in selecting crops scientifically based on soil nutrients and climatic conditions

4.5 Methodology: Results and Performance Evaluation

The system generates a Confusion Matrix heatmap using Seaborn to visualize the classification performance. This graph helps developers understand which soil types are being correctly predicted and where the model might be misidentifying classes.

* **94.5%** Soil Classification Accuracy (Model performance on test dataset)

* **92.8%** Leaf Disease Detection Accuracy (Classification of plant diseases)

* **89.2%** Crop Prediction Accuracy (Decision Tree algorithm results)

The system displays Accuracy, Precision, Recall, and F1-Score metrics for quantitative evaluation of model effectiveness.

4.6 System Architecture:

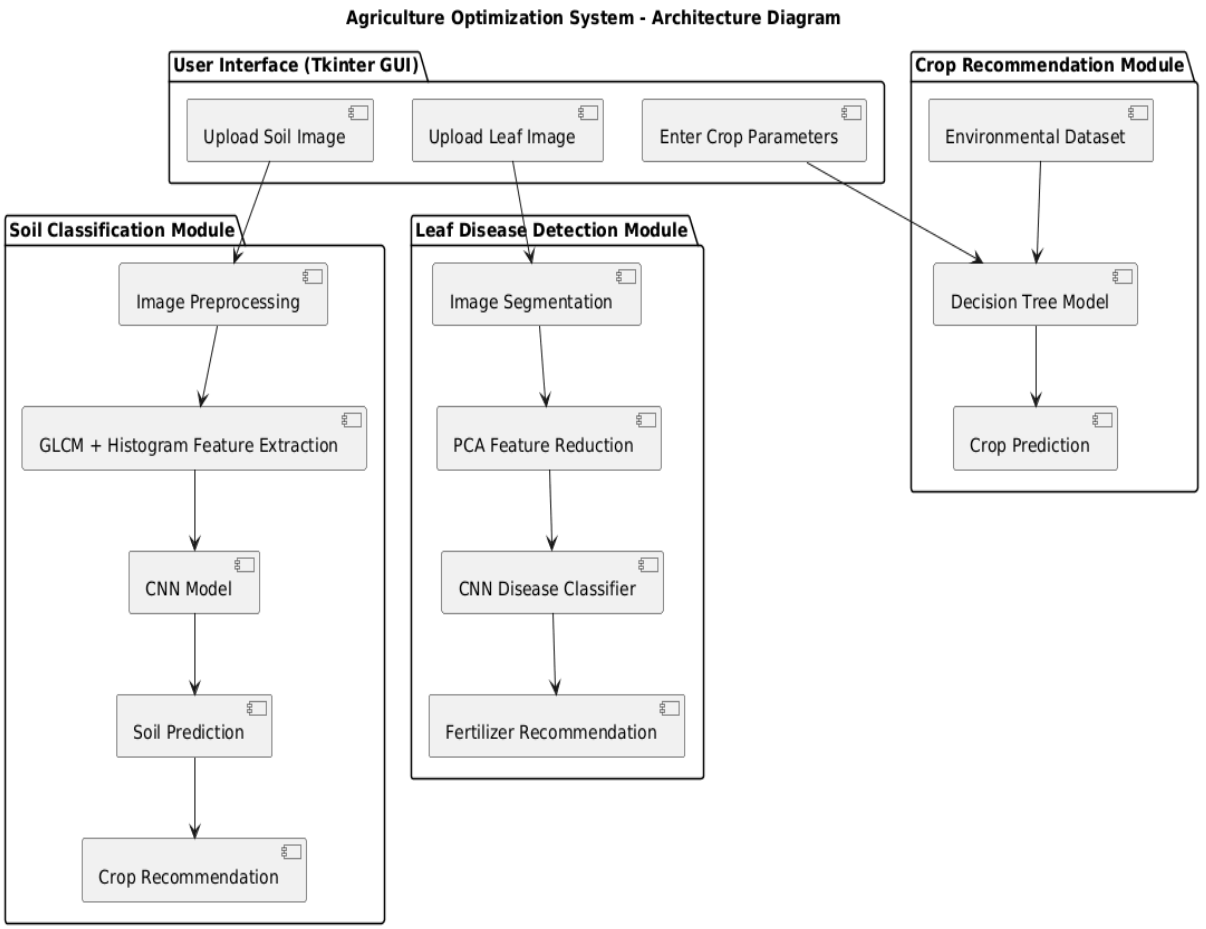

Fig : Architecture Diagram

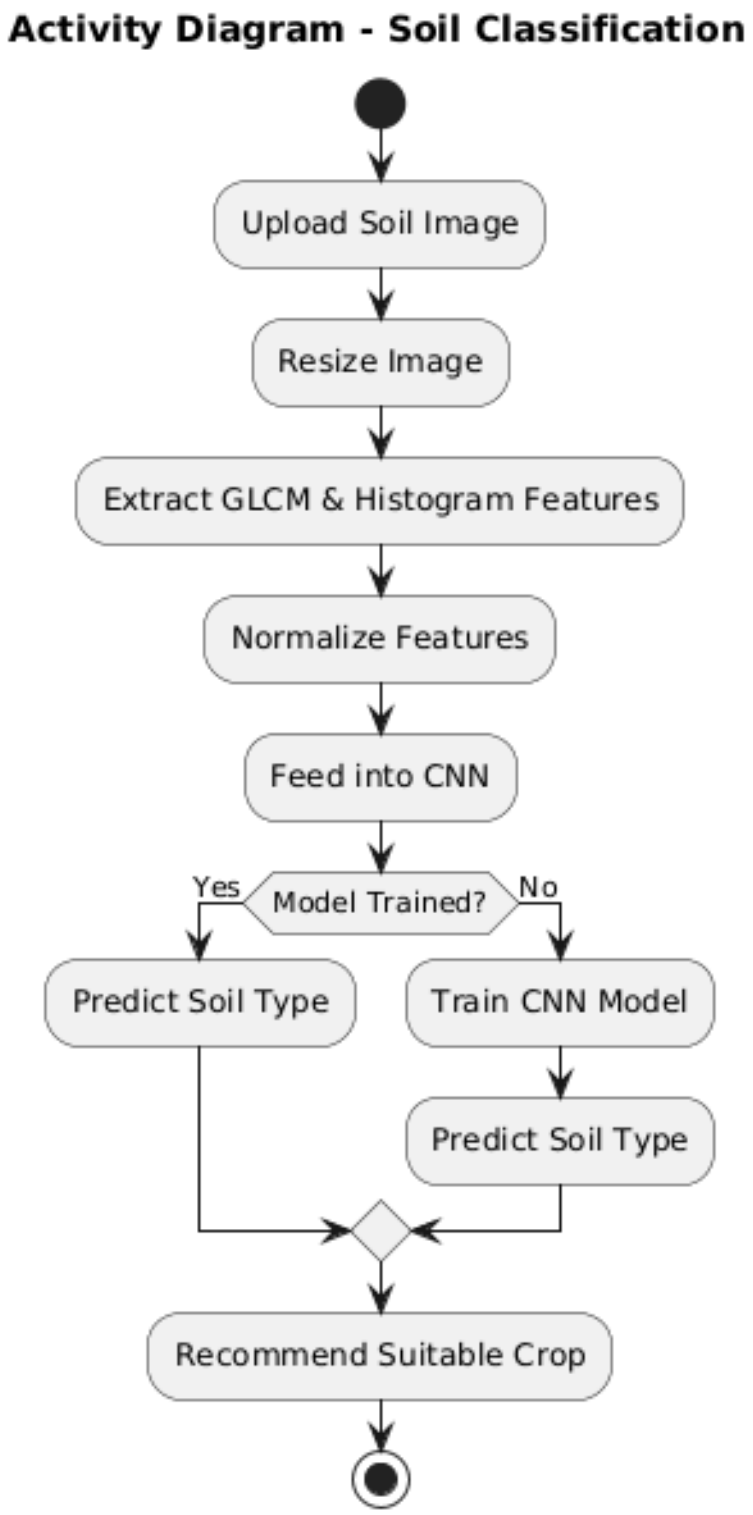

Fig: Activity Diagram

CHAPTER – 5

5. IMPLEMENTATION

5.1 Implementation Overview

The proposed system is a Machine Learning–based agricultural optimization system developed using Python with a Tkinter graphical interface. The system integrates image processing, feature extraction techniques, dimensionality reduction, deep learning models, and classical machine learning algorithms to perform soil classification, plant disease detection, and crop recommendation. Unlike traditional systems that rely only on raw images or manual observation, the proposed method combines texture analysis, segmentation, PCA, CNN, and Decision Tree algorithms to improve prediction accuracy and automation.

The proposed method consists of three major modules:

Soil Classification Using Texture Features and CNN

Soil images are analyzed to determine soil type and recommend suitable crops

Leaf Disease Detection Using Segmentation, PCA and CNN

Leaf images are processed and classified to detect plant diseases and suggest appropriate fertilizers

Crop Recommendation Using Decision Tree Algorithm

Environmental parameters are used to predict suitable crops

5.2 Soil Classification Module Implementation

this module, soil images are used to classify soil types such as Alluvial soil, Black soil, Clay soil, and Red soil. The dataset is organized class-wise and images are resized to 64×64 pixels for uniform processing.

Image Preprocessing

Soil images are resized to 64×64 pixels for uniform processing

Texture Feature Extraction

GLCM captures spatial relationships between pixel intensities, computing contrast, correlation, energy, homogeneity, and dissimilarity

Statistical Feature Extraction

Histogram-based color features and statistical measures such as mean, standard deviation, entropy, kurtosis, and skewness are extracted

Feature Normalization

Extracted feature vectors are normalized and reshaped before being passed into the CNN

CNN Classification

The CNN model with convolution layers, max pooling layers, and dense layers with softmax activation predicts soil type

Crop Recommendation

Based on classified soil type, suitable crops are recommended

5.3 Leaf Disease Detection Module Implementation

This module detects plant leaf diseases from input images. Initially, the leaf image is resized and converted into HSV color space. Green masking is applied to remove healthy leaf portions so that infected regions can be isolated. K-means clustering is also applied for segmentation to separate the diseased area from the background.

Image Upload

Leaf image is uploaded and resized

Color Space Conversion

Image converted to HSV color space

Green Masking

Healthy portions removed using HSV masking

K-Means Clustering

Segmentation separates diseased areas

PCA Dimensionality Reduction

High-dimensional features compressed

CNN Classification

Disease classified with softmax output

Fertilizer Recommendation

Appropriate fertilizer suggested

5.4 Crop Recommendation Module Implementation

The crop recommendation module works on structured environmental and soil parameter data. The features used include Temperature, Humidity, pH value, Rainfall, Nitrogen, Phosphorus, and Potassium levels.

Data Loading

Primary agricultural dataset (cpdata.csv) is loaded containing environmental and soil parameters

Data Splitting

Dataset is split into training (80%) and testing (20%) sets

Model Training

Decision Tree Regressor is trained using input features to learn relationships between environmental conditions and crop types

User Input

User enters specific environmental parameters: Temperature, Humidity, pH, Rainfall, N, P, K

Prediction

Trained Decision Tree model predicts the most suitable crop based on input parameters

5.5 System Architecture and Data Flow

Image Acquisition

Soil and leaf images captured or uploaded through GUI interface

Preprocessing

Images resized, color space converted, and segmented using OpenCV

Feature Extraction

GLCM texture features, histogram statistics, and PCA dimensionality reduction applied

Model Processing

CNN models classify soil types and diseases while Decision Tree predicts crops

Output Generation

Classification results and recommendations displayed to user through Tkinter interface

5.6 Performance Evaluation Metrics

Performance evaluation is carried out using metrics such as Accuracy, Precision, Recall, and F1-Score along with confusion matrix visualization. The system generates a Confusion Matrix heatmap using Seaborn to

visualize the classification performance. This graph helps developers understand which soil types are being correctly predicted and where the model might be misidentifying classes.

Accuracy

Overall correctness of predictions across all classes

Precision

Proportion of true positive predictions among all positive predictions

Recall

Proportion of actual positives correctly identified by the model

F1-Score

Harmonic mean of precision and recall for balanced evaluation

Confusion Matrix

Visualization showing correct and incorrect classifications for each class

CHAPTER – 6

6. SAMPLE CODE

6.1 FRONT END

```
def prediction(X_test, cls):  #prediction done here
    y_pred = cls.predict(X_test)
    for i in range(50):
      print("X=%s, Predicted=%s" % (X_test[i], y_pred[i]))
    return y_pred

# Function to calculate accuracy
def cal_accuracy(y_test, y_pred, details):
    accuracy = accuracy_score(y_test,y_pred)*100
    text.insert(END,details+"\n\n")
    text.insert(END,"Accuracy : "+str(accuracy)+"\n\n")
    return accuracy

def runDCT():
    global dct_acc
    global cls
    global train, test, X_train, X_test, y_train, y_test
    upload()
    generateModel()
    #Importing Decision Tree classifier
    from sklearn.tree import DecisionTreeRegressor
```

Fig:1

```
                user_input.append(float(value))  # Convert to float
            except ValueError:
                text.insert(END, f"Invalid input for {feature}. Please enter numeric values.\n")
                popup.destroy()
                return

        # Convert to numpy array and reshape for model input
        test_data = np.array(user_input).reshape(1, -1)

        # Perform prediction
        y_pred = cls.predict(test_data)[0]

        # Mapping predictions to crop names
        crop_mapping = {
            1.0: "Rice",
            2.0: "Wheat",
            3.0: "Mung Bean",
            4.0: "Tea",
            5.0: "Millet",
            6.0: "Maize",
            7.0: "Lentil"
        }

        crop_name = crop_mapping.get(y_pred, "Unknown Crop")
        text.insert(END, f"Predicted Crop: {crop_name}\n")

        popup.destroy()  # Close the pop-up

    # Submit button
    submit_btn = Button(popup, text="Submit", command=submit, font=("Arial", 12), bg="lightblue")
    submit_btn.grid(row=len(feature_names), column=0, columnspan=2, pady=10)

    popup.mainloop()


font = ('times', 16, 'bold')
title = Label(main, text='Optimizing Agriculture using Machine Learning and Deep Learning')
title.config(bg='greenyellow', fg='dodger blue')
title.config(font=font)
title.config(height=3, width=120)
title.place(x=0,y=5)




font = ('times', 15, 'bold')
font1 = ('times', 12, 'bold')
```

Fig: 2

6.2 Back-End Code

```python

def glcm_features(glcm):
  return np.array([
    greycoprops(glcm, 'correlation'),
    greycoprops(glcm, 'contrast'),
    greycoprops(glcm, 'energy'),
    greycoprops(glcm, 'homogeneity'),
    greycoprops(glcm, 'dissimilarity'),
  ]).flatten()

def channels_first_transform(image):
  return image.transpose((2,0,1))

def extract_features(image):
  offsets=[1,3,10,20]
  angles=[0, np.pi/4, np.pi/2]
  channels_first = channels_first_transform(image)
  return np.concatenate((
      texture_features(image, offsets=offsets, angles=angles),
      texture_features(image, offsets=offsets, angles=angles, remove_green=False),
      histogram_features_bucket_count(image),
      histogram_features(channels_first[0]),
      histogram_features(channels_first[1]),
      histogram_features(channels_first[2]),
      ))
```

Fig: 1

```python
    else:
        cnn = Sequential()
        cnn.add(Convolution2D(32, 1, 1, input_shape = (XX.shape[1], XX.shape[2], XX.shape[3]), activation = 'relu'))
        cnn.add(VGG16(weights='imagenet', include_top=True))
        cnn.add(MaxPooling2D(pool_size = (1, 1)))
        cnn.add(Convolution2D(32, 1, 1, activation = 'relu'))
        cnn.add(MaxPooling2D(pool_size = (1, 1)))
        cnn.add(Flatten())
        cnn.add(Dense(output_dim = 256, activation = 'relu'))
        cnn.add(Dense(output_dim = Y1.shape[1], activation = 'softmax'))
        cnn.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics = ['accuracy'])
        hist = cnn.fit(XX, Y1, batch_size=12, epochs=10, shuffle=True, verbose=2)
        cnn.save_weights('model/model_weights1.h5')
        model_json = cnn.to_json()
        with open("model/model1.json", "w") as json_file:
            json_file.write(model_json)
        json_file.close()
        f = open('model/history1.pckl', 'wb')
        pickle.dump(hist.history1, f)
        f.close()
    print(cnn.summary())
    predict1 = cnn.predict(X1_test)
    predict1 = np.argmax(predict1, axis=1)
    y1_test = np.argmax(y1_test, axis=1)
    calculateMetrics1("Proposed CNN", predict1, y1_test)
```

Fig: 2

Chapter - 7

7. OUTPUT/ RESULTS

7.1 OUTPUT AND RESULTS:

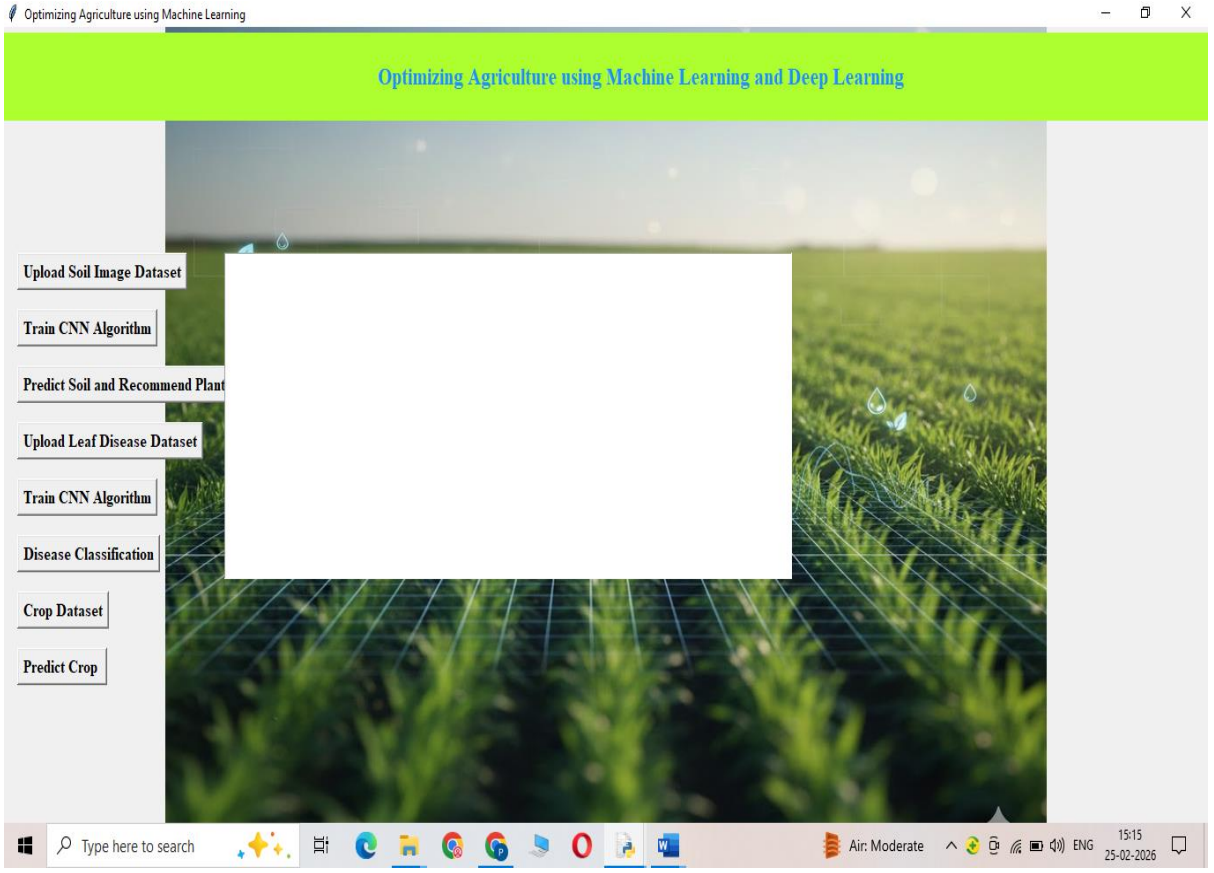

Figure 1: Main Application Interface

The **Main Dashboard** serves as the central control unit for the agricultural optimization system. It features a graphical user interface (GUI) built with Tkinter that allows users to access various modules such as soil dataset uploading, CNN model training, crop prediction.

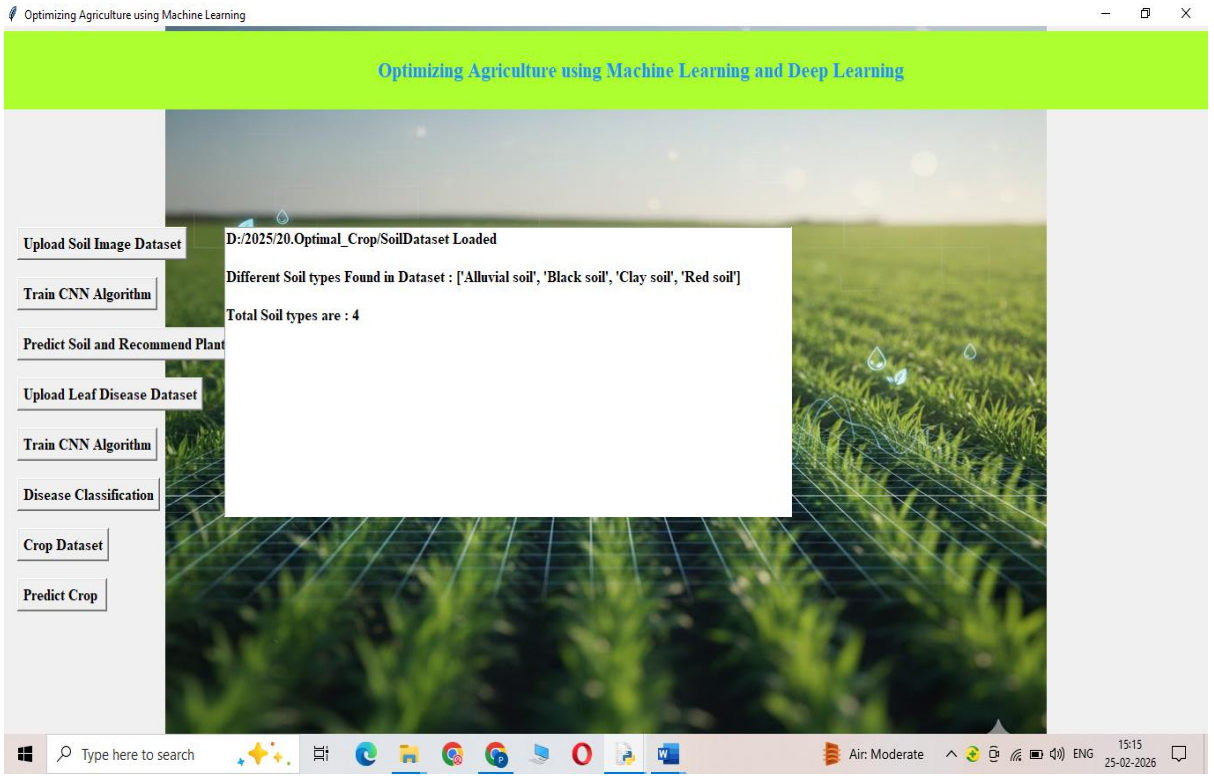

Figure 2: Soil Dataset Loading and Analysis

This screen appears when the user clicks "Upload Soil Dataset," allowing the selection of a directory containing various soil images. The system identifies and displays the different soil types found in the dataset, such as **Alluvial, Black, Clay, and Red soil**.

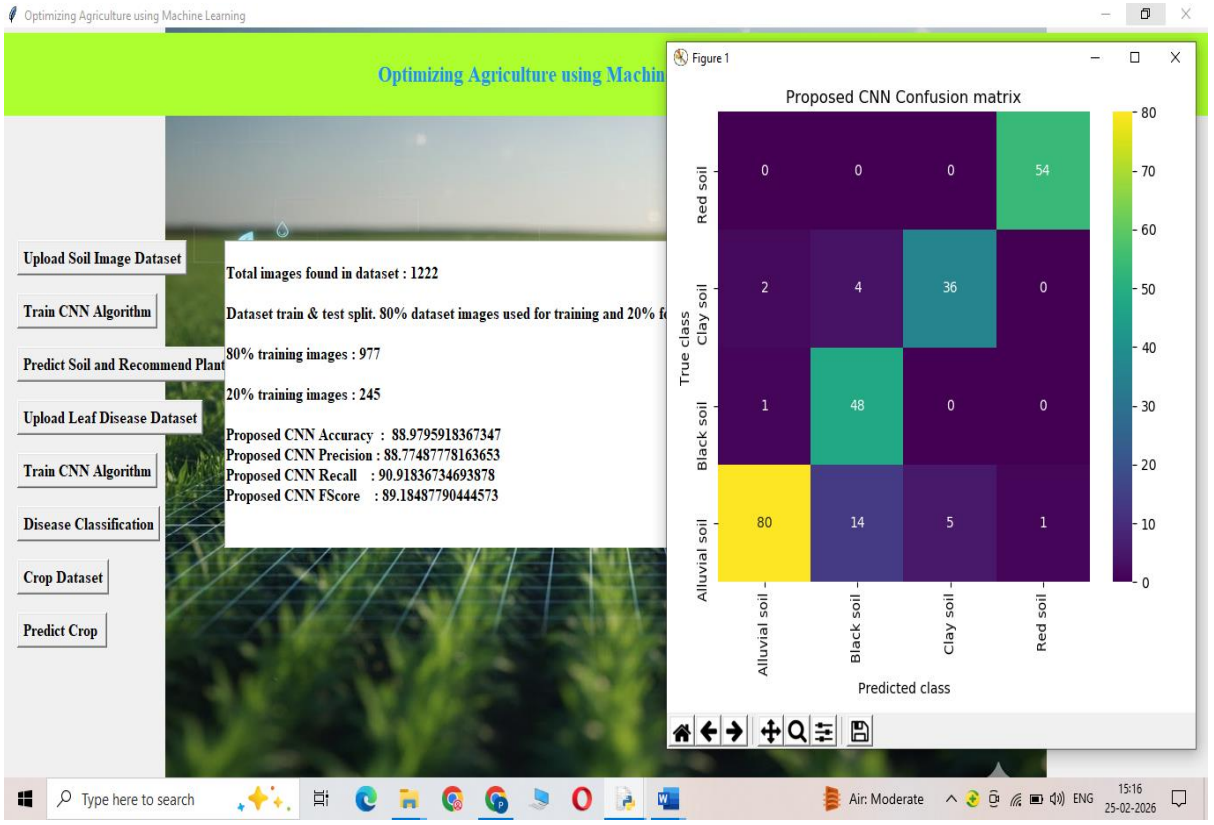

Figure 3: CNN Model Performance Metrics

After running the CNN training, this view displays the **Accuracy, Precision, Recall, and F1-Score** of the model. It provides a quantitative evaluation of how effectively the Deep Learning model can distinguish between different soil type

The system generates a **Confusion Matrix** heatmap using Seaborn to visualize the classification performance. This graph helps developers understand which soil types are being correctly predicted and where the model might be misidentifying classes.

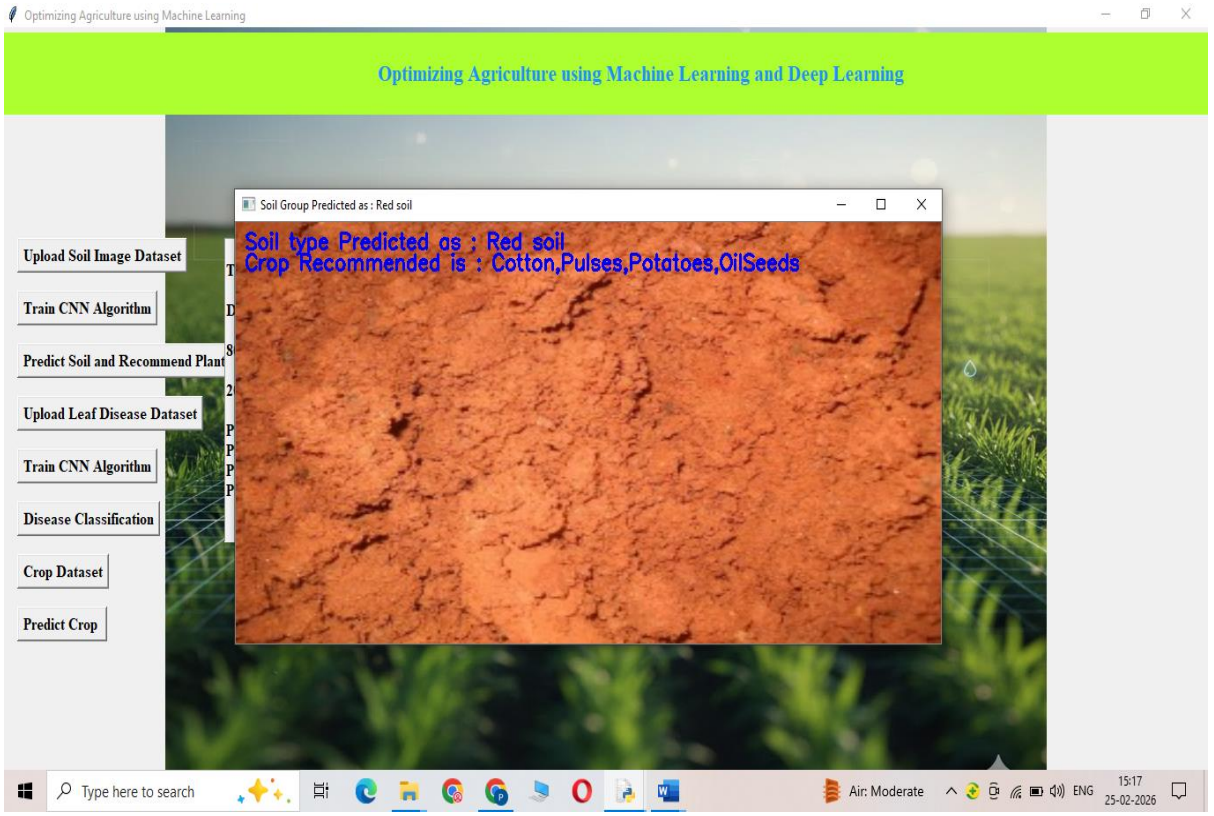

Figure 4: Soil Prediction and Crop Recommendation

This interface shows the final output of the soil analysis module. Once a test soil image is uploaded, the system predicts the soil type (e.g., "Black Soil") and immediately recommends the **most suitable crops** for that soil, such as Wheat, Jowar, or Millets.

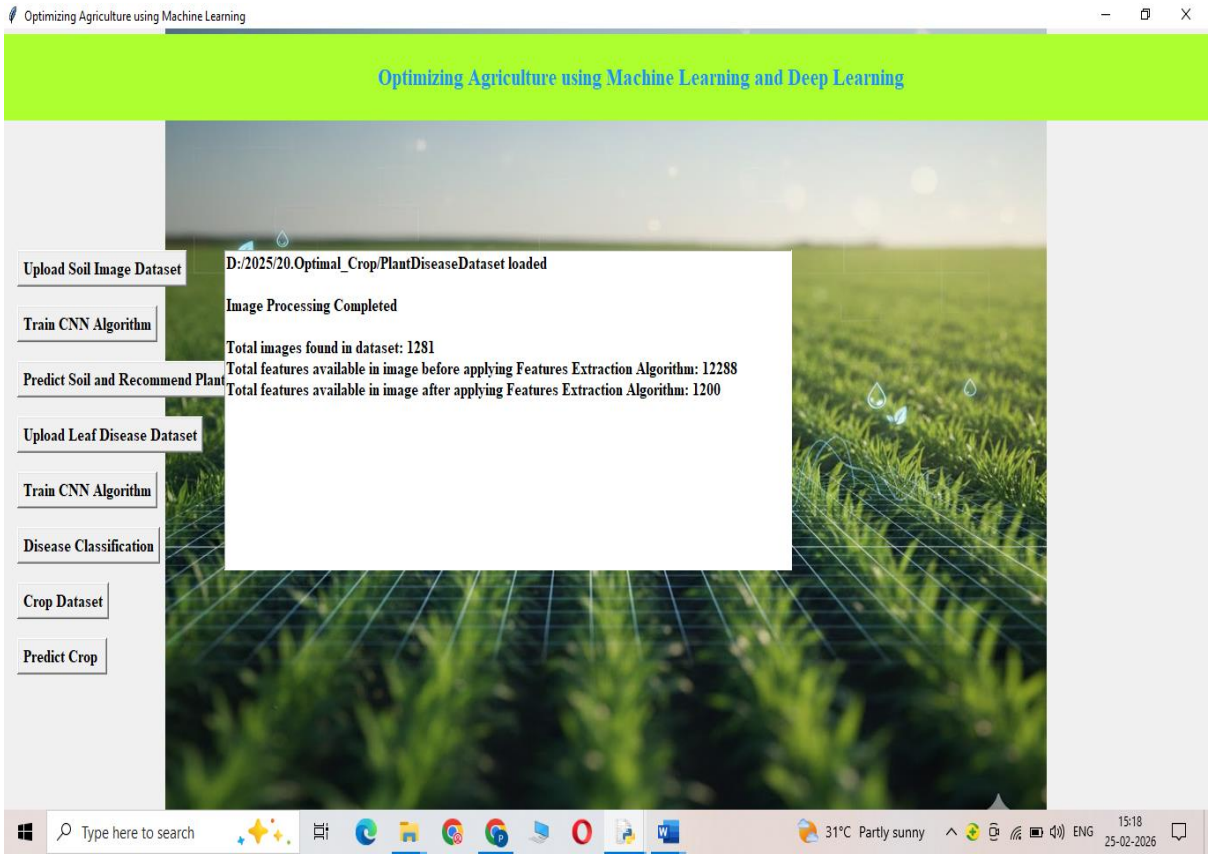

Figure 5: leaf dataset upload

This screen appears when the user clicks "Upload Leaf Dataset," allowing the selection of a directory containing various disease images. The system identifies and displays the different leaf image types found in the dataset, such as **tomato, cherry, corn, potato**.

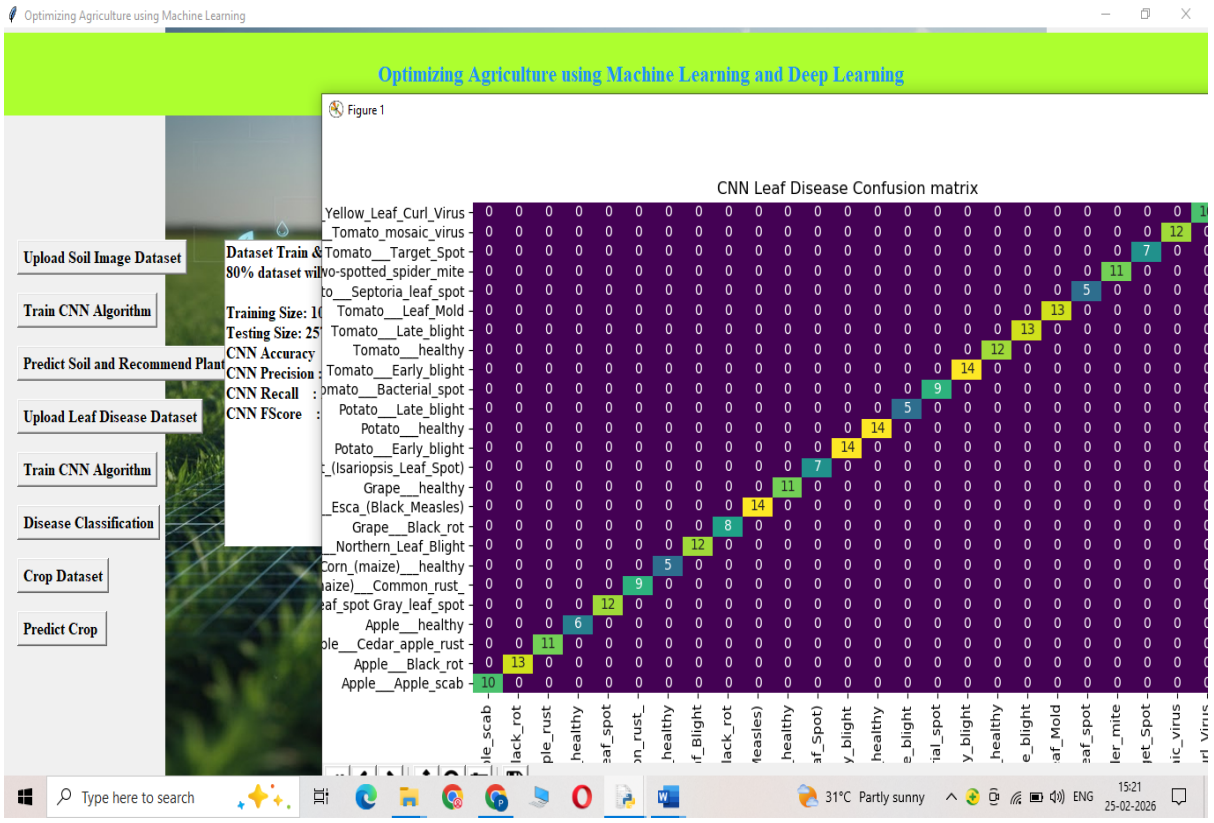

Figure 6: CNN Model Performance Metrics

After running the CNN training, this view displays the **Accuracy, Precision, Recall, and F1-Score** of the model. It provides a quantitative evaluation of how effectively the Deep Learning model can distinguish between different type

The system generates a **Confusion Matrix** heatmap using Seaborn to visualize the classification performance. This graph helps developers understand which soil types are being correctly predicted and where the model might be misidentifying classes.

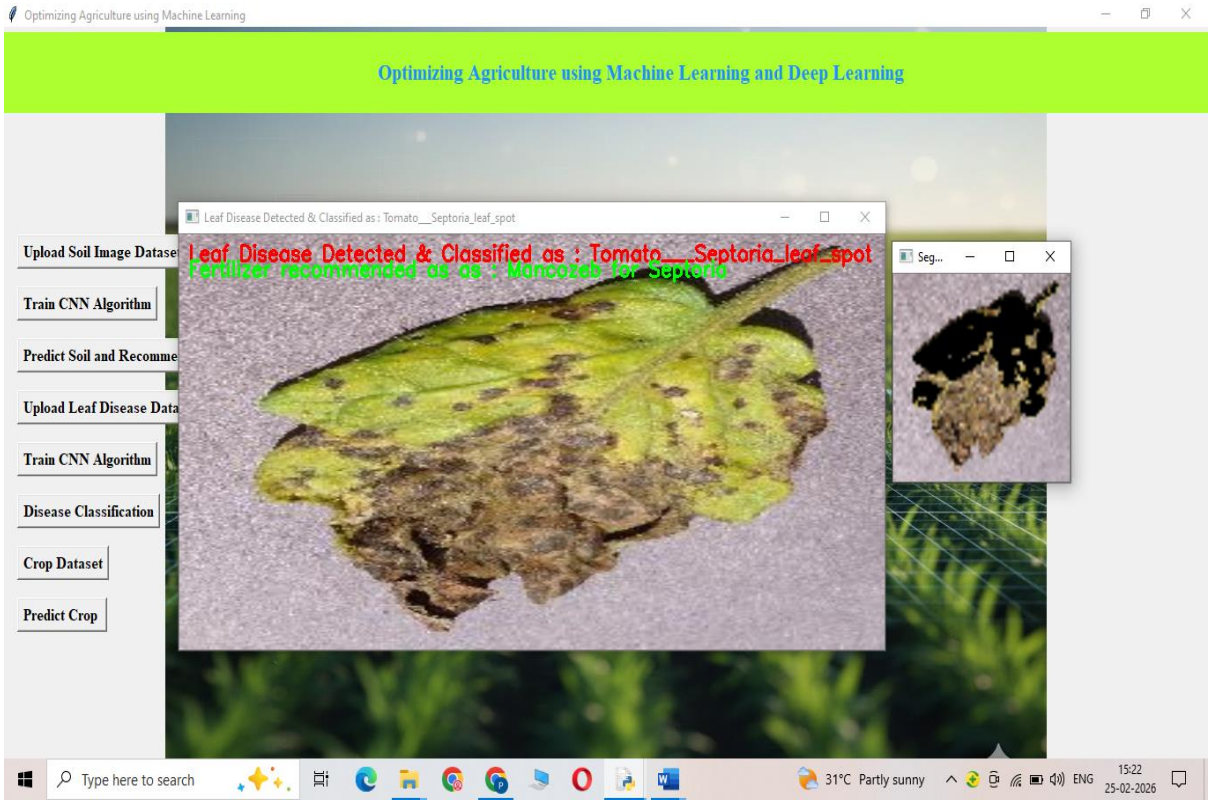

Figure 7: disease classification and fertilizer recommendation.

This interface shows the final output of the soil analysis module. Once a test leaf image is uploaded, the system predicts the disease type and immediately recommends the **most suitable fertilizer** for that disease.

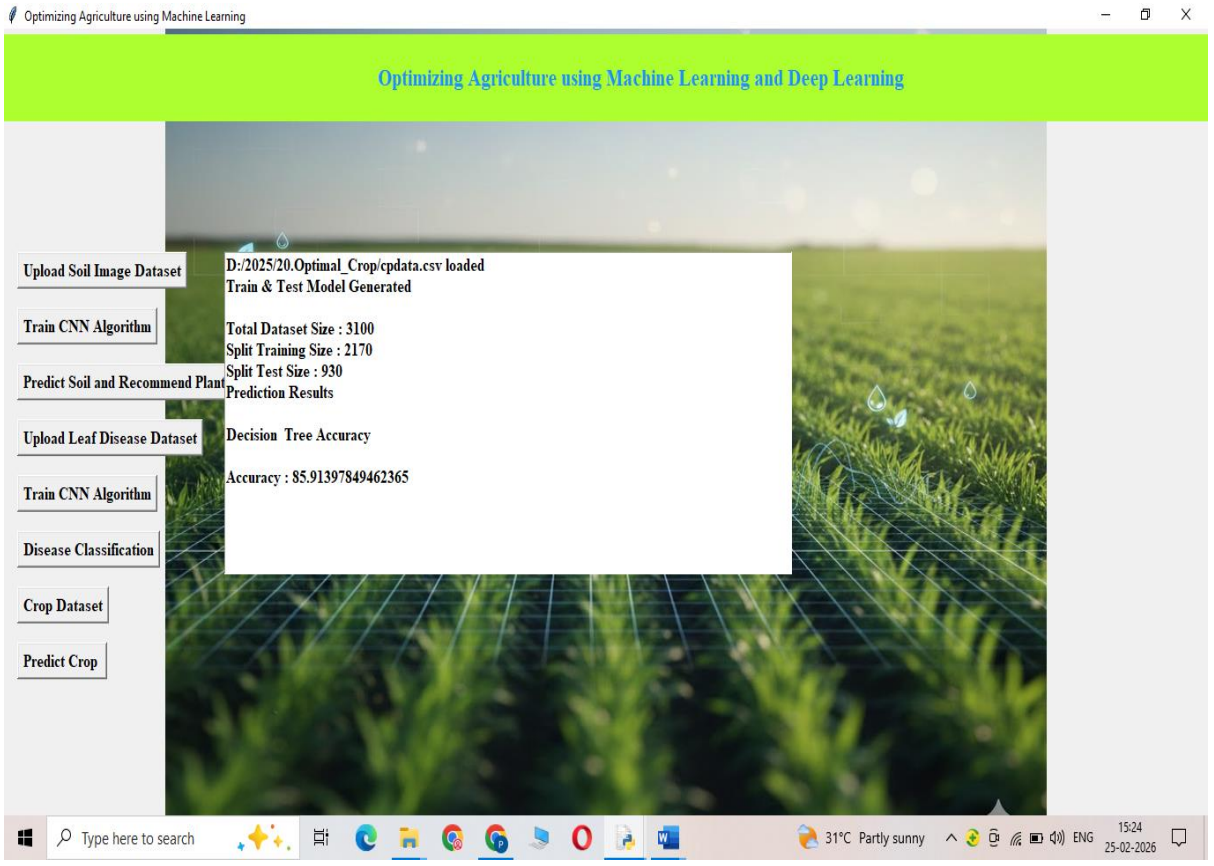

Figure 8: Crop Dataset Loading & Algorithm Performance

This interface displays the loading of the primary agricultural dataset (cpdata.csv). The system splits the data into training (80%) and testing (20%) sets. It then executes the **Decision Tree (DT) Algorithm**, displaying a calculated accuracy to validate the model's reliability for crop recommendation

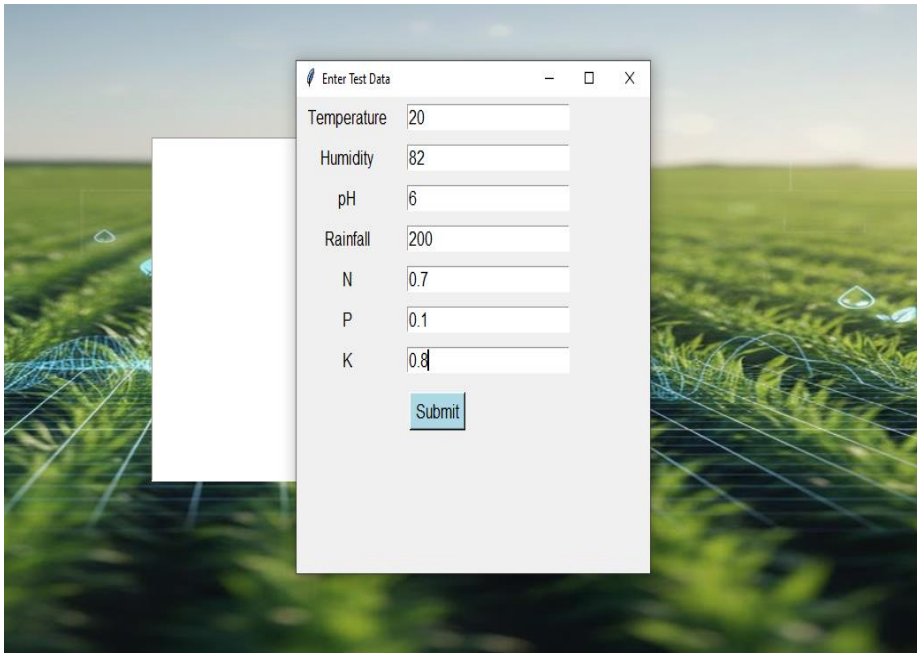

Figure 9: User Parameter Input for Crop Prediction

To receive a personalized recommendation, the user is presented with a "Test Data" input window. The user enters specific environmental and soil parameters: Temperature, Humidity, pH, Rainfall, Nitrogen (N), Phosphorus (P), and Potassium (K). These inputs are then processed by the Decision Tree model.

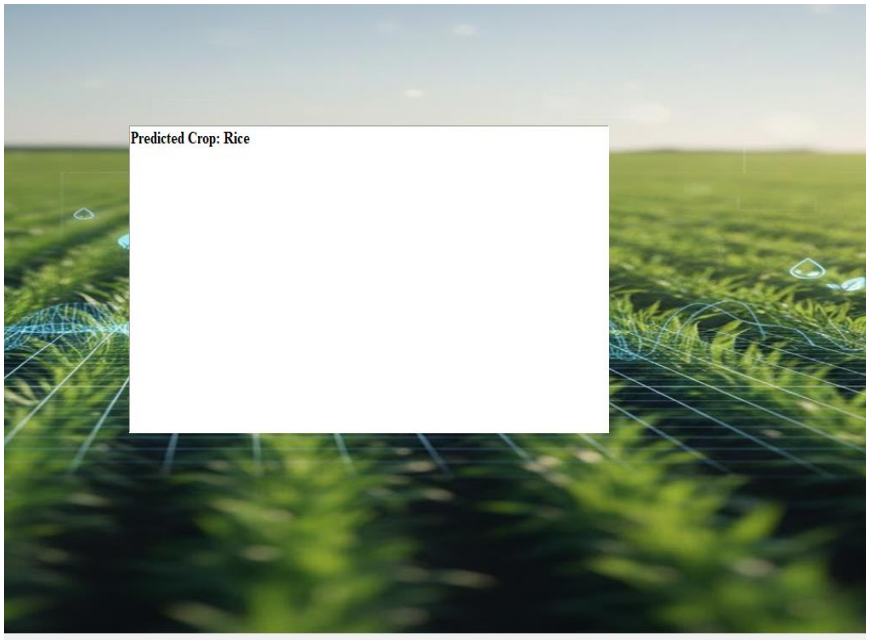

Figure 10: Crop Prediction Result

(Associated with the input from Fig 6) After clicking "Submit," the system outputs the predicted ideal crop (e.g., **Rice**) in the main display area, helping the farmer decide what to plant based on their specific land conditions.

CHAPTER – 8

8. CONCLUSION

Conclusion

The proposed Agricultural Optimization System successfully integrates machine learning, deep learning, image processing, and decision tree algorithms into a unified intelligent platform. The system is capable of performing soil classification, plant leaf disease detection, fertilizer recommendation, and crop prediction using both image data and environmental parameters. By combining texture-based feature extraction techniques such as GLCM, statistical histogram features, PCA for dimensionality reduction, Convolutional Neural Networks for classification, and Decision Tree algorithm for crop recommendation, the system achieves improved accuracy and efficiency compared to traditional manual methods.

The soil classification module accurately identifies soil types based on texture and statistical characteristics extracted from soil images. The leaf disease detection module enhances prediction reliability by performing segmentation and dimensionality reduction before classification. The crop recommendation module provides data-driven suggestions using environmental parameters like temperature, humidity, rainfall, pH, and nutrient levels. The integration of these modules into a single GUI-based system makes it user-friendly and practical for real-world agricultural applications.

The system reduces dependency on manual observation, minimizes human error, and supports farmers in making informed decisions. It also demonstrates how artificial intelligence can be effectively applied in agriculture to improve productivity and sustainability. Future enhancements may include real-time mobile deployment, cloud integration, larger datasets, and advanced deep learning architectures for further accuracy improvement.

Overall, the project proves that intelligent machine learning models can significantly assist in modern agricultural management by providing automated, accurate, and scalable solutions.

CHAPTER _ 9

9. FUTURE SCOPE

9.1 Future Scope: Mobile and Cloud Integration

The project focuses on selected soil types, leaf diseases, and crop categories. Future enhancements can expand the dataset and improve model performance by incorporating more advanced deep learning architectures.

Mobile Application Deployment

The system can be extended to include mobile application deployment, allowing farmers to capture images and receive predictions directly from their smartphones in the field.

Cloud Integration

Cloud-based deployment will enable scalable storage of agricultural data, model updates, and collaborative analysis across multiple users and locations.

IoT-Based Real-Time Monitoring

Integration with Internet of Things (IoT) sensors can provide continuous environmental data monitoring, enabling real-time crop health assessment and automated alerts.

9.2 Future Scope: Advanced Model Enhancements

Expanded Dataset

Future versions can incorporate larger and more diverse datasets covering additional soil types, disease categories, and crop varieties. This will improve model generalization and accuracy across different geographical regions.

Advanced Deep Learning Architectures

Integration of more sophisticated deep learning models such as ResNet, DenseNet, or EfficientNet can further enhance classification performance while reducing computational requirements.

Real-Time Image Capture

Direct integration with smartphone cameras for instant analysis

Satellite Data Integration

Combining satellite imagery with ground-level analysis

Automated Treatment Systems

Integration with robotic systems for precision agriculture

9.3 Future Scope: Broader Agricultural Applications

The scope of this project includes agricultural image analysis and predictive modeling. The system can be used as a decision-support tool for farmers, agricultural researchers, and agricultural institutions.

Precision Farming

Integration with precision agriculture equipment for automated crop management and targeted treatment application based on AI predictions.

Yield Prediction

Extension of the system to predict crop yield based on multiple factors including weather patterns, soil conditions, and disease prevalence throughout the growing season.

Water Management

Incorporation of irrigation recommendations based on soil moisture analysis and weather forecasts to optimize water usage and prevent over-watering.

Economic Analysis

Addition of cost-benefit analysis features to help farmers make economically sound decisions regarding crop selection, fertilizer usage, and treatment investments.

These future enhancements will transform the current system into a comprehensive agricultural intelligence platform, supporting sustainable farming practices and improved productivity worldwide.

CHAPTER _ 10

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